EVENTUAL PERIODICITY OF LINEAR GENERALIZED STONE–WALES PATHS IN AFFINE FULLERENE FAMILIES WITH CONVEX LATTICE-FACE DECOMPOSITIONS

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Abstract. We study literal linear generalized Stone–Wales paths in fullerene duals. Our main result concerns an integer affine family admitting a fixed decomposition into embedded convex lattice polygons with sides in the six triangular-lattice directions. With fixed gluing and affine corner coordinates, we give a uniform reduction of literal-path existence to a fixed monadic second-order sentence over unary finite words, using maximal face passages, their two-row outputs and all vertex collisions. Eventual periodicity on every eventually valid integer affine ray then follows from the known equivalence between MSO and regular languages. This is a geometric definability reduction, not a new general periodicity principle. We record the resulting ray statements for eleven specified catalogue families, alongside closed conditions on their stated domains. Existing Lean results establish arithmetic periodicity of several family predicates and determine the decisions on two example rays for every value of the ray parameter. For K3, a graph level sufficient condition is formalized on an explicit quotient specification. These are separate from a formal proof of the general geometric theorem. Human proof peer review and novelty remain open; the general argument is a received independently derived prose proof with AI review, and no effective MSO compiler is supplied.

Editorial and evidence status. This public-release English edition continues Japanese v3. The main result is its convex-face Theorem 1a and catalogue Corollary 1b. The wider fixed-oblique-direction argument is retained only as a proof sketch in Section 8, outside the abstract's claims. The contribution is the uniform geometric reduction to a fixed MSO sentence on unary words; the final periodicity implication is known. Section 9.3 separates that reduction from standard MSO, parametric Presburger and fixed-boundary iteration results. The independent verifier R004 derived the convex-face proof without reading the research team's proof, construction or code. The statement was known. Disclosures about O2 mapping metadata and D12 single-point information are retained; strict blindness is not claimed. One AI coordinator reviewed the route. Translation and packaging are not a third mathematical derivation or human peer review. The author has approved the authorship, affiliation, licenses and AI-use statement for this edition. The first paper's A3 classification is a separate work; its 816-class census is not reproduced here. A new D12 classification is not incorporated.

1. Definitions, signatures and valid rays

1.1. The literal path.

Definition 1. In the dual G of a fullerene, a literal linear gSW path is a sequence $p_0, \dots, p_e$ with odd $e \geq 3$, all vertices distinct, consecutive and distance-two vertices adjacent, $\deg p_0 = \deg p_e = 5$, and $\deg p_1 = \deg p_{e-1} = 6$.

Machine-readable claim: `def:literal`.

This is Definition 1(a) of [1]. Internal degree-five vertices are allowed. Even vertex count is part of the definition, not a new obstruction. Positive means existence of such a path; no chemical barrier, radial operation, or operation permitting heptagons is inferred. A fullerene is a simple

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three-connected cubic plane graph with only pentagonal and hexagonal faces. Its dual is a spherical triangulation with twelve degree-five vertices and all other degrees six.

When every three-cycle is facial and an edge has just the two common neighbours supplied by its incident triangles, an oriented triangle (a, b, c) has a unique next vertex: the common neighbour of b, c different from a. A simple path is a prefix of this forced sequence. If ρ is its first repeated-vertex index, a legal endpoint must occur strictly before ρ, not at ρ.
Machine-readable claim: `prop:forced`.

The conditional finite-graph statement is formalized as `R002.finite_literal_path_decision`. Its hypotheses and fixed start must be supplied by geometry. The noncomputable construction is not a certified executable Python or C classifier. For fullerene duals, the nonfacial-three-cycle exclusion in the received prose route uses the established cyclic edge-connectivity result cited in [3]; the original proof of that external result was not rechecked by the editor.

1.2. **Signatures and eventual validity.** *Machine-readable claim:* `def:signature`.

Work in the triangular lattice $\Lambda = \mathbb{Z}^2$, with $R(x, y) = (-y, x + y)$. A signature specifies a fixed finite set of polygon charts, paired boundary segments, corner equivalence classes, and transition maps $z \mapsto R^j z + d$. Corner coordinates and translations are integer affine functions of the parameters; rotations and gluing combinatorics are fixed. Along a ray $p(t) = at + b$, $t \in \mathbb{N}$, we require eventually positive edge lengths, embedded chart interiors, only the specified seam identifications, noncrossing macro edges, and a valid simple fullerene dual. Every cone has a marked corner label. Seams need an ownership convention, and full corner equivalence classes are used. Positive parameters and positive area alone do not prove validity.

A component $a_i t + b_i$ eventually lies in the positive integers exactly when $a_i > 0$ or $a_i = 0, b_i \geq 1$. In the first case an entry bound is $\max(0, \lceil (1 - b_i)/a_i \rceil)$. Replacing 1 by 2 gives the analogous bound for a lower limit two. A negative slope has no infinite positive tail. These arithmetic entry bounds do not certify the quotient.

2. The main result: convex lattice-face decompositions

Theorem 1a (Convex lattice-face rays)**.** *For all sufficiently large integers t, suppose a valid simple fullerene dual has a decomposition into a fixed finite number of embedded closed convex lattice-polygon disks. Every side has positive length and one of the six lattice directions. Straight angles and auxiliary degree-six corners are permitted. No distinct boundary points of one disk are identified. Corner coordinates and transition translations are integer affine in t, while rotations, side pairings and corner identifications are fixed. All degree-five vertices occur among the marked corners, with a fixed label set. Then the set of t for which a literal path exists is eventually periodic. The conclusion also holds when these hypotheses hold separately on finitely many residue classes modulo a fixed integer.*

Machine-readable claim: `thm:aligned`.

Thus some $t_0 \geq 0, P \geq 1$ satisfy the same decision at t and $t + P$ for all $t \geq t_0$. The received theorem proves existence. No practical threshold, least period, implemented MSO-to-DFA compiler, or Lean proof of this geometric statement is supplied. Section 3 gives the received route; its original proof, corrections and coordinator review are fixed in the packet. The hypotheses describe the objects to which it applies, not a generic validity-testing algorithm.

Corollary 1b (Specified catalogue families)**.** *On every integer affine ray eventually remaining in the valid positive integer domain of each of A1, A2, B2, C1, K3, E10, D12, D14, D1 and O2, literal-path existence is eventually periodic. The same conclusion holds for A3 on rays eventually satisfying $k > l > 0$, by its separate classification and arithmetic argument.*

Machine-readable claim: `cor:catalog`.

There are eleven families in this statement. Positive boundaries with a parameter fixed at 1 are included, notably K3 and O2 and E10 with $r = 1$. Collapsed zero boundaries are not. The corollary gives no two-variable closed classification merely from ray periodicity. A3 generally has changing parent-edge directions and is not silently placed under Theorem 1a. The catalogue terminology and corrections are Graver's [2]; the constructions are not claimed as new. Additional leapfrog families are discussed only as prospects in Section 8.

3. Proof of the convex-face theorem

This section presents R004's second derivation on the common six-direction domain. It does not use the uncompleted oblique-direction transfer conditions. Its arithmetic-to-word reduction is specific; arbitrary Presburger relations are not claimed to be finite-word MSO.

3.1. **Maximal face passages and their outputs.** *Machine-readable claim:* `lem:rows`.

Rotate a passage direction to $(1, 0)$. A suitable phase of the forced vertex sequence is

$$q_{2j} = (j, h), \qquad q_{2j+1} = (j, h + 1). \tag{1}$$

A convex six-direction polygon is specified by

$$L_x \le x \le U_x, \qquad L_y \le y \le U_y, \qquad L_s \le x + y \le U_s.$$

Its bounds are affine in t after the finitely many max/min orders stabilize. Redundant collinear sides do not alter this description. With $L_y \le h \le U_y - 1$, the triangle $T_i = \{q_i, q_{i+1}, q_{i+2}\}$ lies in the polygon precisely in the ranges

$$\begin{aligned} A_e &= \max(L_x, L_s - h), & B_e &= \min(U_x - 1, U_s - h - 1), \\ A_o &= \max(L_x, L_s - h - 1), & B_o &= \min(U_x - 1, U_s - h - 2). \end{aligned} \tag{2}$$

Even indices are $2j$ for $A_e \le j \le B_e$, odd indices are $2j + 1$ for $A_o \le j \le B_o$. Put $\delta = A_e - A_o$ and $\epsilon = B_o - B_e + 1$, both in $\{0, 1\}$. The full triangle-index set is empty or the interval

$$\ell = 2A_e - \delta \le i \le u = 2B_e + \epsilon. \tag{3}$$

This includes the one-odd-triangle case $A_e = B_e + 1$, $\delta = \epsilon = 1$. It must not be discarded as an empty even range.

The first two vertices of an entered passage already belong to the previous passage. Its new output is only $q_{\ell+2}, \dots, q_{u+2}$, comprising

$$\begin{aligned} y = h: &\quad A_e + 1 \le x \le B_e + 1, \\ y = h + 1: &\quad A_e + 1 - \delta \le x \le B_e + \epsilon. \end{aligned} \tag{4}$$

For example the upper row has $x \ge L_x + 1 - \delta$, $x + y \ge L_s + 2 - \delta$, $x \le U_x - 1 + \epsilon$, and $x + y \le U_s + \epsilon$. The last constant was corrected by hand in the received proof, followed by a new finite check; the check did not originally discover that error. Empty rows are allowed, but a nonempty passage has nonempty output.

Root and terminal passages can be truncated. There are finitely many marked cone corners and at most ten oriented initial triangles at each. The initial triangle fixes h and an index of the form twice an affine expression plus fixed parity. Store its first two vertices separately. An endpoint corner and choice of output row similarly determine the last index. Truncation adds affine bounds on x. Both cuts are made when the whole path lies in one passage. Internal degree-five corners are not compulsory stopping points. The two endpoint neighbours must be distinct from all degree-five labels.

3.2. **States, transitions and all collisions.** *Machine-readable claim:* `lem:restricted`.

A state is (τ, h), where the finite type τ records face, direction, max/min branches, parity, subdivided entry/exit side, and root/terminal information. A constant C bounds $|h| \le Ct + C$. The number of states or face passages need not be bounded independently of t. The relevant relations are Boolean combinations of

$$\pm h \le at + b, \qquad \pm h \pm h' \le at + b, \qquad at + b \ge 0, \tag{5}$$

and fixed-modulus congruences. A relation such as $2h + h' \le at + b$ with distinct free variables is not part of this language. Substituting one variable for another can be reduced by fixed parity and rounding.

Within a fixed type, each endpoint's three normal coordinates has the form $c(t)$, $h + c(t)$ or $-h + c(t)$. When the last ordered pair of a passage is glued to the next one's first ordered pair, the directions are rotated together. In direction-aligned coordinates the linear part of this transition is the identity, so $h' = h + c(t)$, not a doubling map. The paired unit edge determines the opposite face; its interior contains no macro corner. The conditions therefore have form (5).

Each output row fixes one of $x, y, x + y$ at $\pm h + c(t)$ and imposes affine bounds in all three normals. For distinct row normals the determinant is ± 1: their intersection is integral, and the third normal is their sum or difference. Substitution into both rows gives (5). Equal normals first require equal values; the remaining integer interval is nonempty exactly when every lower bound is at most every upper bound. The same restricted form results. Between different faces, compare through every shared side and every common marked corner label. Rotations permute the three normals with signs. Embeddedness and fixed gluing ensure these exhaust quotient identifications, even when faces share several sides. Auxiliary flat corners are included. Comparison with the root's two saved vertices uses the same formulas. Since an embedded face row is injective, pairwise disjoint outputs and their disjointness from the root are exactly the full vertex-distinctness condition.

3.3. **Folding the restricted arithmetic into a finite word.** *Machine-readable claim:* `lem:folding`.

For $t \ge 2$, write a height variable uniquely as $h = jt + y$, $0 \le y < t$. The quotient j ranges over a fixed finite set. Set $t = 2n + \eta$, $\eta \in \{0, 1\}$, and split y into

$$L : y = v, \qquad R : y = t - 1 - v \quad (0 \le v < n), \qquad M : y = n \quad (\eta = 1). \tag{6}$$

Original type, j and $L/R/M$ are finitely many copies of the positions of a unary word of length n. A middle copy is placed at the first position and interpreted as coordinate n, independent of that position.

For two ordinary positions the inequalities become

$$\alpha v + \beta w \le 2cn + d, \qquad \alpha, \beta \in \{-1, 1\}.$$

The coefficient of n is even. If $\alpha = -\beta$, then for $c \ne 0$ truth stabilizes, while $c = 0$ is order with a fixed successor offset. If $\alpha = \beta = 1$, the sum lies between 0 and $2n - 2$. The only nontrivial cases are $c = 0$, a finite list near the left end, and $c = 1$. For $c = 1$, putting $a = n - 1 - v, b = n - 1 - w$ gives $a + b \ge -d - 2$, either automatic or the exclusion of finitely many near-end pairs. The case $\alpha = \beta = -1$ is its integer complement. If one position is a middle copy, only one variable remains and the nontrivial inequalities are fixed distances from an end. Two middle copies give an affine truth value that stabilizes.

Fixed-modulus congruences are represented by cyclic MSO colours on the word: the first position has colour zero and each successor increments the colour. The final colour determines n modulo the fixed modulus. Finite copies of element or set quantifiers are expanded copy by copy. Thus every fixed Boolean combination used above is MSO-definable. This argument does not permit an arbitrary Presburger relation such as $v + w \le n$; it uses the even coefficient obtained by the folding step.

3.4. **The exact path sentence and unary regularity.** *Machine-readable claim:* `lem:mso-path`.

Choose a nonempty state set S, root a and terminal b. Terminal states have no outgoing transition and root states no incoming one. Record the parity of the number of vertices already output as a finite type. Initially it is even, since the root has two saved vertices; each transition adds the parity of $u - \ell + 1$. Impose the following MSO conditions:

(1) If $a = b$, $S = \{a\}$. Otherwise a has in/out degrees $(0, 1)$, b has $(1, 0)$, and every other state has $(1, 1)$ within S.
(2) Every nonempty proper subset of S has a transition, in either direction, across its boundary. This is unary-set connectivity.
(3) Outputs of distinct states are disjoint, and all outputs are disjoint from the two saved root vertices.
(4) Endpoint degrees are $5, 6, 6, 5$ as in Definition 1. Total vertex count is even and at least four.

For one state, the last size condition is $u - \ell + 1 \geq 2$; with two or more nonempty-output states it is automatic. Degree conditions are membership or nonmembership in the finite marked-cone set. No binary relation quantifier is used.

On a finite set, the first two conditions give one directed path through all of S: a reached cycle would create an extra incoming edge, and a separate cycle would contradict connectivity. Join the passages in this order. Ordered seam pairs give consecutive and skip adjacency; the third condition gives distinct vertices. Size, parity and degree conditions give exactly the literal path.

Conversely, split any literal path into maximal consecutive passages in its convex faces, truncating only at its ends. Equations (2)–(4) recover their states. Nonempty output and path simplicity prevent a repeated state. Any extra eligible successor within S would share the next output vertex with the actual successor, contradicting disjointness. Hence the selected states satisfy all four conditions. One-passage paths and paths with internal cones are included in both directions.

Finite-word MSO defines exactly the regular languages [4, Section 3.1, Theorem 3.2]. Apply this established result only at this last logical step. For each η, a deterministic automaton over one letter eventually repeats a state, so its accepted lengths n are eventually periodic. Combine $t = 2n + \eta$ and, if necessary, the initially fixed residue branches. This proves Theorem 1a. The geometric reduction belongs to the received R004 proof, not to the cited automata theorem. Neither the full proof of that external theorem nor an MSO compiler was newly verified by this manuscript's editor.

4. Applying the theorem to the specified families

Machine-readable claim: `prop:family-decompositions`.

The applicability evidence is R004's fixed family-decomposition report and its coordinator review. D12 uses two subdivided hexagons and D14 two convex hexagons whose long sides are subdivided with lengths r, s, r. O2 uses two hexagons and six triangles. K3 uses two caps, each divided into four triangles, and three hexagons. These decompositions retain the positive parameter-1 boundaries.

D1's original thirty-degree sides do not directly satisfy the six direction hypothesis. Cut a triangle $A_iB_iW_i$ from each oblique side and move it to the back by the same lattice isometry. For example,

$$A_0 = (r, 0), \quad B_0 = (r + p, p), \quad W_0 = (r, p), \quad C = (-p, r + 2p),$$

and $W_0 - C = \alpha(A_0 - C) + \alpha(B_0 - C)$ with $\alpha = (r + p)/(2r + 3p)$ and $0 < 2\alpha < 1$. The pieces are contained and disjoint. The result is eight convex pieces: one regular hexagon, six quadrilaterals, and the back hexagon. New corners are flat degree six. Area equality alone is not used to infer the same quotient.

For E10 put $m = 2r + s$, tube circumference $C = 3m$. Move the longitudinal cut from (p, p) to $(0, 2p)$, obtaining three lattice parallelograms of width $m < C$ and two caps: five disks, also when $r = 1$. Cut-corner positions involve p mod C. If C grows, $\lfloor p/C \rfloor$ eventually stabilizes; if C is

constant, use fixed residue classes. Delete duplicate auxiliary corners and fix their order to obtain affine coordinates and fixed gluing. This does not classify E10 with $r = 1$.

Family and domain	Received reason for all valid affine rays
A1, $r \geq 1$	Integer GC absence, including the separate $r = 1$ case; eventual period one.
A2, $p \geq 1$	Nonconstant rays eventually give nonprimitive GC images; constant rays are constant graphs.
B2, $p, r \geq 1$	Its closed condition is a finite disjunction of affine equalities.
C1, $a, b \geq 1$	Its closed condition is $b = 1$.
A3, $k > l > 0$	Separate complete classification and arithmetic; Section 6.
K3, $r, s \geq 1$	Closed-condition arithmetic for $r, s \geq 2$; convex decomposition includes parameter 1.
E10, $p, r, s \geq 1$	Arithmetic for $r \geq 2$; the five-piece decomposition includes $r = 1$.
D12, D14, D1	The convex decompositions on valid positive integer parameters.
O2, $t, d \geq 1$	Convex decomposition, including parameter 1; absence is received only for $t, d \geq 2$.

This proves Corollary 1b using the stated geometric receipts and the separate arithmetic results. It does not cover all 112 catalogue entries or every zero-boundary degeneration.

5. Closed conditions and the scope of partial results

The following are received prose classifications of valid catalogue graphs. Independent derivation of the same conclusion is distinct from checking every coordinate convention in the research team's tables. Lean arithmetic for a predicate does not by itself prove its equivalence to a graph property.

5.1. **The specified closed domains.** *Machine-readable claim:* `prop:closed`.

Family/domain	Literal path condition	Received scope
A1, $r \geq 1$	Never positive	Integer GC for $r \geq 2$; at 1 no degree-six vertex. R004 report 4 and GC proof.
A2, $p \geq 1$	$p = 1$	Existing C_{60} witness and integer GC. R004 report 14 does not supply a new $p = 1$ witness.
C1, $a, b \geq 1$	$b = 1$; IPR iff $a, b \geq 2$	R004 report 3 and C1 proof; coordinator review of the team's argument.
B2, $p, r \geq 1$	$p = 1$ or $\lvert p - r \rvert = 1$	R004 report 4 and B2 proof, whole positive domain.
K3, $r, s \geq 2$	$s \equiv \pm 1 \pmod{r}$	R004 positive and necessary prose proofs. Lean graph sufficiency only; arithmetic rays formalized.
E10, $p, s \geq 1, r \geq 2$	$m = 2r + s$ and $p \equiv d \pm 1 \pmod{m}$ for some $d \in \{0, r, r + s\}$	R004 reports 5–6 and both prose directions. Predicate-ray arithmetic formalized; full graph iff not formalized.

K3 writes $s = qr + u$: $u = 0$ is excluded by integer GC, $2 \leq u \leq r - 2$ by belts of widths $r, u, r - u$, and $u = 1, r - 1$ by explicit simple paths. The overlapping small cases $r = 2, 3$ are separate. The iff is not extended to $r = 1$ or $s = 1$. E10's candidate congruences come from the positions of corners around the tube; reaching a cone does not alone establish simplicity. The research team's full table of eighteen E10 regions, walls and witnesses is not all independently reviewed.

5.2. **Partial branches and boundary examples.** *Machine-readable claim:* `prop:partial`.

O2 is received as absent and IPR for $t, d \geq 2$, with the mapping-metadata disclosure retained. For D14 and D1 with both parameters at least two, the research team has belt-based absence drafts; R004's all-ray periodicity receipt is not a new independent proof of that closed absence claim. Common integer scaling by at least two gives absence in the specified leapfrog domains, but does not classify their primitive points.

For E10$(p, 1, s)$, $m = s + 2$, the research team's witnesses give sufficiency when $p < m$ or $p \bmod m \in \{0, 1, 2, m-2, m-1\}$. General necessity is not proved. Among the 84 positive (p, s) with $N = 12 + 2s^2 + 12s + 12p(2 + s) \leq 740$, exactly E10$(9, 1, 4) = C_{740}$ is negative; all 83 smaller examples are positive. E10$(17, 1, 11)$ and $(31, 1, 7)$ give further negatives at C_{3038}, C_{3542}. These three examples are non-IPR. R004's report 4 independently constructed and checked them. This is minimality only inside the $r = 1$ E10 family, not a general iff.

The pre-existing D12 branches are retained as historical scope: $\gcd(r, s) \geq 2$ gives absence; $r \geq 3$, $s = qr \pm 1$, $q \geq 1$ gives existence by the separately derived simple witnesses. The research team's $|r - 3s| = 1$ and $r \geq 4s$ branches have their earlier review limitations. A full D12 classification, including newer work, is outside this paper and this packet.

6. Arithmetic on affine rays and exact examples

6.1. **The determinant reduction.** *Machine-readable claim:* `lem:divisibility`.

Let $A(t) = \alpha t + \beta$ be eventually positive and $B(t) = \gamma t + \delta$. A constant A gives a fixed-modulus congruence. If $\alpha > 0$, then

$$A \mid B \quad \Longrightarrow \quad A \mid \alpha\delta - \gamma\beta. \tag{7}$$

For nonzero right side, divisibility is impossible once $A > |\alpha\delta - \gamma\beta|$. For zero right side, the ratio $B/A = \gamma/\alpha$ is constant and is either always integral or never integral. Finite Boolean combinations of affine divisibilities are eventually periodic; a growing modulus gives eventual constancy. Take a common multiple of fixed moduli and a maximum stabilization threshold. This elementary arithmetic is not proposed as new number theory. The received Lean files establish the exact predicates below.

For all integer ray coefficients, define parameters by $(at + b).\,\mathrm{toNat}$. The formal statements for all rays use *domain and positive predicate*; they do not call out-of-domain graphs positive. On an eventually positive domain the conversion agrees with the integer affine parameter.

Machine-readable claim: `thm:k3-arithmetic`.

`Claude2.k3_ray_all` proves eventual periodicity of K3Dom $\wedge$ K3Pos on all integer affine rays, where the domain is $r, s \geq 2$ and the predicate is $r \mid (s - 1)$ or $r \mid (s + 1)$. If r is nonconstant on a valid tail, `k3_ray_const` gives eventual constancy. If $r = b \geq 2$ is constant, `k3_ray_period` gives period b from any specified domain-entry threshold. These need not be least periods.

Machine-readable claim: `thm:e10-arithmetic`.

`Claude2.e10_ray_all` proves the corresponding all-ray statement for domain $p, s \geq 1, r \geq 2$ and the six divisibilities in the E10 condition. If $2r + s$ is nonconstant, the predicate is eventually constant; when it is constant, `e10_ray_period` gives that positive modulus as a period on the given domain tail.

Machine-readable claim: `thm:other-arithmetic`.

`b2_ray_all`, `c1_ray_all`, `a1_ray_all` and `a2_ray_all` give eventual constancy of their domain-qualified predicates. The conjunction/disjunction/negation closure and domain qualification are proved in the same arithmetic layer.

6.2. **A3 is a separate variable-direction case.** *Machine-readable claim:* `thm:a3-arithmetic`.

For $k(t) = at + b, l(t) = ct + d$ set $\Delta = ad - bc$. If $\Delta \neq 0$, $\gcd(k, l) = \gcd(k, l, \Delta)$, so primitivity is periodic modulo $|\Delta|$. When $\Delta = 0$ and the ray is nonconstant, it is eventually an increasing

integer multiple of a fixed primitive direction and hence negative by integer GC. Constant rays are constant. In the primitive case put $m = \min(l, k-l)$. The A3 criterion is $m = 1$, or quotient $q = \lfloor k/m \rfloor \geq 3$ with $m \mid (k \pm 1)$, or $q = 2$ with $r = k - 2m = 1$ or $r \mid (m \pm 1)$. Branches and quotient stabilize, including an integer limiting ratio by checking the sign of the constant remainder. The persistent $q = 2, r = 0$ growing branch is not primitive. Apply (7) to the remaining tests.

The first paper proves the full graph classification. Its `ClaudeCor.criterion_ray`, restated as `Claude2.a3_ray`, formalizes the arithmetic existence of a threshold and positive period. A separate release bridge gives graph-path periodicity. Neither result certifies an executable threshold/period algorithm. R004's separate arithmetic argument permits period $|\Delta|$ for nonconstant $\Delta \neq 0$ rays and period 1 otherwise. This is not a least-period statement or formalization of the general geometric theorem. After classification, eventual periodicity is also a consequence of [5, Theorem 1.15] and is not a separate novelty claim.

6.3. **Four previously solved rays.** *Machine-readable claim:* `ex:exact-rays`.

Ray, $t \geq 0$	Positive values	Valid period and threshold
$\mathrm{D12}(124 + 3t, 41 + t)$	All t	$P = 1, t_0 = 0$
$\mathrm{D12}(125 + 3t, 41 + t)$	None	$P = 1, t_0 = 0$
$\mathrm{K3}(43 + 2t, 170 + 5t)$	$40, 42$	$P = 1, t_0 = 43$
$\mathrm{E10}(200 + 5t, 17 + 2t, 23 + t)$	$3, 10, 17, 34$	$P = 1, t_0 = 35$

The original compressed-word calculations did not use the known family congruences. Later independent prose arguments recover all t from independently derived family conditions, or from explicit D12 blocks. The expected statements were known, not blind predictions.
Machine-readable claim: `ex:k3-formal`.

`Claude2.k3_example` now proves for every $t \in \mathbb{N}$ $\mathrm{K3Pos}(43 + 2t, 170 + 5t) \leftrightarrow (t = 40 \vee t = 42)$.
Machine-readable claim: `ex:e10-formal`.

`Claude2.e10_example` proves the corresponding E10 predicate iff $t \in \{3, 10, 17, 34\}$. The determinant constants used are $123, 127$ for K3 and $142, 144, 739, 749, 681, 691$ for E10. Divisibility bounds t before the remaining cases are checked by kernel arithmetic. These are all-t arithmetic proofs, not graph iff proofs by themselves.

The earlier D12 two-ray receipt covers every $s \geq 41$: $\mathrm{D12}(3s+2, s)$ is negative and $\mathrm{D12}(3s+1, s)$ has a path with $56s + 12$ distinct vertices. Its reported computer-assisted proof checks 2,058 nonnegativity conditions and 756 no-integer-solution conditions, 2,814 in total. A separately coded reread by the same verifier is not a further reviewer. The coordinator's independent construction at $s = 58$ agrees, with a 3,260-vertex positive witness; it does not audit all 2,814 conditions. These records are neither Lean proofs nor a full D12 classification.

7. Received formalization and what it does not cover

7.1. **K3 graph sufficiency.** *Machine-readable claim:* `thm:k3-sufficient`.

The completed declaration and its type are

`Claude2K3.k3_sufficient` :
`Claude2Aligned.K3Sufficient`.

For natural $r, s \geq 2$, it proves that $s \equiv \pm 1 \pmod{r}$ implies an actual literal path in the quotient graph

$$\texttt{Claude2Aligned.k3Graph}(r, s).$$

The specification uses eleven convex polygons, equivalence closure of explicit affine side gluings, and adjacency from unit differences inside a common chart. The literal path and actual neighbour-set degree definitions are those of the first paper's specification.

The proof follows R004's tube construction. A glued-site invariant is $(\mathrm{hgt}, \mathrm{ang} \bmod 6r)$. It is preserved by all 42 directed edge gluings. It is not globally injective: the two central cap triangles are not separated by it. On tube points with horizontal coordinate difference at most one, however, equality of quotient vertices forces equality of points. The explicit paths are

$$P_n^+ = (-(n \bmod 2), \lfloor (n+1)/2 \rfloor), \qquad P_n^- = (n \bmod 2, \lfloor n/2 \rfloor),$$

with $4s-2$ vertices for $s = qr+1$ and $4s+4$ for $s = qr-1$. Chart membership, consecutive and skip adjacency, distinctness, and the four endpoint degree requirements are proved on the quotient. The invariant is not used to claim a global vertex normal form.
Machine-readable claim: `lem:k3-corners`.

`Claude2K3.deg_corner` proves degree five for all twelve labelled corner representatives. It does not by itself prove they exhaust all degree-five vertices or all global vertex types.
Machine-readable claim: `lem:k3-hex-interior`.

`deg_hex_interior` proves degree six in hexagon interiors.
Machine-readable claim: `lem:k3-tri-interior`.

`deg_tri_interior` proves degree six in the six noncentral cap-triangle interiors. General seam interiors and the central cap triangles are not covered by that partial degree classification. No K3 graph-necessity proof was received at the fixed source checkpoint used here. The prose iff in Section 5 is not upgraded to a formal iff.

The specification was written by the same worker as these K3 proofs. Human interpretation against the catalogue remains outstanding. The first paper's owner's check of its A3 challenge statement does not apply to this K3 specification. Shared specification files contain unused D12 definitions, but no new D12 classification is a root or claimed result of this package.

7.2. **Historical conditional kernels.** *Machine-readable claim:* `lem:historical-kernels`.

The earlier research layer supplies conditional pieces: `finite_literal_path_decision` assumes a finite type and two common neighbours; GC lattice transport excludes odd endpoints in a scaled lattice; the four cylinder arithmetic declarations prove the height-kernel displacement with stated divisibility hypotheses; `induction_uniform_depth` assumes a finitely branching child list and genuinely decreasing lexicographic rank. The concrete `ietComponentChildren` models only removal of equal last labels, not every subtraction, equal-length elimination or accelerated suffix. The geometric hypotheses, all-start coverage, full compressed action and a certified program are not supplied by those conditional types. There were historically 57 such research declarations; the former total 98 also counted 41 A3 arithmetic entries. Neither count is a formalization percentage or the current library size.

7.3. **Pins, receipt scope and proof dependencies.** The received toolchain is Lean 4.35.0 with the fixed mathlib manifest. The new arithmetic and K3 sufficiency reports record only `propext`, `Classical.choice`, `Quot.sound` as transitive axioms. We supply their exact source closure, types and fixed reports. This edit has not run Lean, rebuilt dependencies, or checked those new declarations in an external kernel. A report of axiom output is not relabelled as fresh raw stdout. Older kernel outputs are retained separately. The first paper's six-kernel Comparator receipt concerns its A3 classification, not this article's general theorem or the new K3 proof.

The claims' expository dependencies and Lean's imported proof-term dependencies are different. A prose statement can remain open for its coordinate or human-semantic bridge while a separate exact formal declaration is proved. Conversely the proved arithmetic predicates do not supply their graph iff interpretation. No human peer review of the proofs is recorded and novelty is not established.

8. An extension to fixed oblique directions (proof sketch)

Machine-readable claim: `claim:oblique`.

Status. The coordinator has reviewed the skeleton of AN Sections 2–6 as a plausible proof sketch. Explicit Section 5 transfer conditions and numerical validation remain incomplete; no independent second proof covers this wider domain. Four obligations below remain. This section is not part of the main theorem or the abstract's result.

The proposed extension replaces the six side directions by an arbitrary fixed finite set of primitive lattice directions, keeping eventual validity, ownership and fixed affine gluing. For a direction v, use transverse coordinate $h = \det(v, z)$. Across a seam the shift is affine. Let D be a common multiple of the nonzero $|\det(v, e)|$ and put $Q = 4D$. Add the transverse cell modulo Q and t modulo Q to the finite labels. Entrance and exit indices are $k_{\text{in}} = \lceil \xi_{\text{in}} \rceil$, $k_{\text{out}} = \lceil \xi_{\text{out}} \rceil$ with $k_{\text{in}} \le k < k_{\text{out}}$. Reordering cells $n = \sigma + Qq$ conjugates the discrete midpoint permutation; it does not claim an order-preserving conjugacy of the continuous intervals.

The AL/AM/AH proposal keeps a full partition into return towers, including saved periodic components. Equal final labels remove a periodic tower; equal lengths merge and remove a label without losing an orbit. Repeated winner cycles are compressed as powers. The claimed bound is on the number of Letter/Cat/Power nodes, not on expanded passage count. Base widths and exponents are affine on fixed residue branches; word lengths and displacement vectors are rational polynomials in t. A lexicographic measure uses leading coefficient sum in the original parameter, number of labels, and constant length; it is not silently reused after coefficient-amplifying reparameterization.

For local ownership, close the side directions under rotations. A unit triangle meets at most $\sum_e (\max_\Delta \det(e, z) - \min_\Delta \det(e, z) + 1)$ candidate macro-edge levels. Embedded noncrossing edges and finitely many corner displays bound the length of the local gluing word. A finite number of charts alone does not bound how often they can be revisited. Entrance-closed, exit-open ownership excludes repeated representations of one position from genuine quotient collisions. Keep the full corner class and the degree-six second-point test; non-IPR graphs can have fewer than 120 legal starts.

Tower return restores midpoint and direction; two returns restore left/right phase. Nonempty output, initial suffix and terminal prefix must be connected before using this as a finite first-repeat bound. Positions are leaf types and ancestor-power repetition indices. A free index j contributes j times a displacement depending only on t. Products jt or jt^2 are allowed coefficients in one-parameter Presburger arithmetic; products of two freely quantified repetition indices are not used. Fix one start address θ throughout:

$$\begin{aligned}\mathrm{Bad}(t, \theta, e) \iff \exists \xi, \eta\, [D_\theta(\xi) \wedge D_\theta(\eta) \wedge \xi \le e \wedge \eta \le e \\ \wedge \neg I_\theta(\xi, \eta) \wedge G_\theta(\xi, \eta)].\end{aligned}$$

Success would be $\exists \theta, e\, [\mathrm{Legal}(t, \theta, e) \wedge \neg \mathrm{Bad}(t, \theta, e)]$. Legal includes odd endpoint index, minimum length, degrees and the preceding output position. Allowing each collision a different start address would be incorrect.

The remaining obligations are: (i) write the first-crossed-edge, containment and order inequalities of the Section 5 transfer, with an oblique-family example; (ii) verify the AH/AM affine tower coefficients for the Q-split IET; (iii) keep effective threshold/period computation dependent on the effective elimination result [6], whose full proof was not reaudited; (iv) compare an oblique-family ray's predicted tail against full graph decisions. None is newly resolved here. BGW supplies eventual periodicity only after the geometric formula has been established.

8.1. **Prospective applications only.** C2 and K4 are specified leapfrogs of C1 and K3; the identified O1$(p, 0, s)$ subfamily is a specified leapfrog of O2. The local coordinate map $(x, y) \mapsto (x - y, x + 2y)$ has determinant 3. These are prospects through the extension, not families in Corollary 1b. Leapfrog is not assumed to preserve absence in general, and the full three-parameter O1 family is not classified. These prospects do not cover arbitrary signatures or every degenerate boundary. For the prospective signatures, twice area is $24a^2 + 48ab + 12b^2$ for C2 and $42r^2 + 36rs$

for K4. Their $a = 0$ and $r = 0$ cases collapse faces, while $b = 0$ and $s = 0$ give zero sides. The zero middle coordinate in the identified O1 subfamily denotes a separately identified leapfrog signature, not permission to discard a zero edge in the old signature.

9. Fixed-width structure, finite checks and novelty

9.1. **Belts and a fixed cylinder.** *Machine-readable claim:* `prop:cylinder`.

Developed triangle midpoints lie on a straight line; vertices alternate on its two neighbouring integer-height rows. Under $z' = Mz + q$, a height $h(z) = n \cdot z + e$ transforms as $n' = nM^{-1}$, $e' = e - n' \cdot q$. If every start is covered by a return preserving position, direction and primitive height, no cone lies inside a band, and all widths are at least two, the required pair of endpoint rows cannot both occur. Width one alone is not sufficient: simplicity and endpoint-neighbour degrees remain necessary. For integer $\mathrm{GC}(k, 0)$, $k \geq 2$, the odd-point determinant ± 1 contradicts membership in the k-multiple lattice; rotations about parent vertices preserve that scaled lattice.

For the fixed-cylinder argument AD let circumference vector $W = gu$, $g = \gcd(|W_1|, |W_2|)$, $h(z) = \det(u, z)$, and insertion vector D have $H = h(D) > 0$. Let L be the least common multiple of the nonzero $|h(v)|$ over the six unit directions. An entrance-state period is

$$Q = 2gL, \qquad \frac{Q}{h(v)}(Hv - h(v)D) \in \mathbb{Z}W. \tag{8}$$

Directions with $h(v) = 0$ loop in the flat core and repeat without reaching a cone. If the caps and collars have a fixed finite vertex set C, at most $|C| + 1$ new cap visits occur before stopping. Overlap of the same three-vertex display is not a new visit. On a residue branch the passages have form $a_i(n) + jv_i$, $0 \leq j \leq \ell_i(n)$, with affine a_i, ℓ_i and fixed v_i, W. All collisions are $a_i(n) + jv_i = a_k(n) + qv_k + zW$, with distinct-position and range conditions. Ordinary Presburger arithmetic then gives periodicity. Q is an entrance period, not necessarily the final or least period.

The uniform cap argument does not extend to growing width/caps. The old $\mathrm{D12}(3, 3t + 4)$ example has passage word $(BA)^{t+1}$ and first legal endpoint index $30t^2 + 88t + 59$. Its $2(t + 1)$ passages are unbounded. The former general bounded-passage forecast was withdrawn.

9.2. **Dated finite evidence.** *Machine-readable claim:* `sec:finite`.

The packet retains, rather than reruns, the relevant evidence. T0770 compares 1,809 arithmetic values on nine known A3 rays; T0771 adds fourteen graph decisions on an independent constructor. Those fourteen are not unseen tests and the 1,809 values are not counted twice. T0887 checks fixed tail coefficients beyond 64 on those rays. T0888 audits E10's fixed descriptor: eight charts, 36 inverse gluings, 108 intervals, 156 connections, 120 descriptors and twelve labels. T0889 audits 71 charts, 342 edges and 21 single-zero boundaries. These are finite checks of supplied data, not the general ownership theorem. The recorded AM audit has 6,564 inputs and 67,312 states; the AN phase audit has 48 directions, 14,904 entrances, 4,005 intervals (1,013 empty), 5,496 lifts and 40,380 conjugacy cells.

R004's report-14 finite arithmetic check has 378 coefficient representations and 13,430 pairs, maximum recorded period 44 and threshold 69. The report-15 checks include 7,056 strips, 2,255 nonempty row outputs, 16,212 subintervals, 99,225 ordered row pairs and 509,160 folded-point comparisons. Their hand-found row correction and earlier type/key failures are kept with their chronology. Numerical agreement does not replace either general proof or repair its hypotheses. No graph construction, R004 checker, or historical scientific audit was rerun for this translation.

The K3 worker's earlier finite mirror agreed with its specification and the verifier's construction on the stated known samples. The two constructions and generated Lean text are separate from the kernel proof. Generated degree formulas were fitted from sample data, but the received Lean theorem checks their universal statements; samples alone would not justify sufficiency. The worker's mirror is not a new independent reviewer of its own specification.

9.3. **Novelty and remaining obligations.** *Machine-readable claim:* `sec:novelty`.
Immediate known consequences. Once a fixed sentence ψ_η describes success on a^n, $t = 2n + \eta$, MSO/regular-language equivalence gives a finite automaton; its unary state orbit is eventually a cycle [4, Section 3.1]. The HTML numbering is Theorem 3.2; the same v1 PDF calls it Theorem 14. Finite fixed-modulus residue branches preserve eventual periodicity. Likewise, a fixed-size parametric Presburger description gives the conclusion directly by [5, Theorem 1.15]. Neither theorem builds the geometric description: an unbounded list of path vertices is not a fixed list of quantified variables, and variable-variable products are not permitted by that Presburger theorem.

For a fixed-width construction $G_n = C[F^n(B)]$ with fixed finite boundary, literal-path existence has a fixed CMSO_2 description. Quantify a vertex set X, an edge set E' and distinct endpoints a, b: (X, E') is connected, has degree one at a, b and degree two elsewhere; $|X| \geq 4$ is even; a, b have ambient degree five and their path neighbours ambient degree six; every two-edge path in E' has adjacent outer vertices in the ambient graph. This permits interior cones and extra chords. The selected subgraph is precisely a literal path, in both directions. Finite-state recognizability for a fixed HR signature [8, Chapter 1, Theorems 1.12 and 1.21(2)] then makes iteration of F a finite-state orbit. Thus this fixed-width existence statement is a short known consequence, not a new principle. Mere bounded treewidth of an arbitrary indexed sequence is insufficient: the uniform repeated construction is essential. The explicit Q, cap-visit bound and actual period-calculation procedure require separate evaluation.
Nearby results and the missing translations. Behmaram–Boutillier's fixed-circumference matching transfer matrix has 2^m states [9, Section 3, Eq. (3.1)]; increasing width does not leave the same matrix, and its observable is matchings, not literal paths. Fischer–Makowsky prove periodicity or semilinearity of model-size spectra for CMSOL classes with bounded width [10, Theorems 1.4, 1.7 and 1.10]. Such a spectrum asserts existence of some model of a size, rather than deciding a specified G_t. Kotek–Makowsky's labelled (bi-)iterative graph families and MSO polynomials admit finite matrix transfers and C^2-finite recurrences [11, Definitions 17–20, Lemma 28, Theorems 30 and 32]. Their bounded-clique-width framework does not automatically include growing lattice patches; a recurrence for values is not itself Boolean periodicity. In each case the geometric family and its parameter require a uniform interpretation satisfying the cited hypotheses.

The syntax $\pm x \pm y \leq c$ is already the octagon domain [12, p. 1]. Bozga–Iosif–Konečný's difference, octagonal and finite-monoid affine relations have acceleration and Presburger transitive-closure results [13, Definition 1 and Sections 4.2.1–4.2.2]. These principally iterate a fixed relation and allow linear drift; here the relation itself varies with t, and all previously output surface vertices must remain distinct. A fixed-dimensional translation preserving that condition has not been supplied. The term "zigzag automata" there does not identify the subject with fullerene zigzags.

Jonoska–Krajcevski–McColm study counter languages of walks returning to a unit cell in periodic digraphs [14]; Wanke studies path/cycle algorithms on finite periodic graphs [15]. For these two sources (S2191, S2192), only the official abstracts and bibliographic records were obtained. We do not infer their full hypotheses or claim an application from an unread theorem. Walks or decidability alone do not preserve literal simplicity or prove periodicity of the parameter set.
Unary length is essential. The standard WS1S encoding of Presburger integers uses their binary representations [16, printed pp. 11–12]. Binary regularity does not imply eventual periodicity of the represented integers: the regular language 10^* represents powers of two. Our reduction uses n as unary word length, not the binary digits of t. A separate formula for each t would also be insufficient.
Boundary of the contribution. The received R005 comparison read the theorem statements and specified sections listed in the packet's source records, not all proofs. Within that reading, no prior result giving the full claim from the same geometric hypotheses was located. Novelty is not established. The remaining candidate contribution is the uniform, two-way translation of geometric passages, seam ownership, legal truncation and all surface-vertex collisions into one fixed MSO

sentence, even with growing widths and passage counts. Fixed-width theory and the known logical conclusions do not supply that translation by themselves; we do not prove that a shorter existing reduction is impossible. The general convex-face proof is received research work, not an already published theorem cited as external authority. Human proof peer review and full comparison with catalogue/zigzag theory remain necessary. The extension's unfinished geometric details do not disappear on receipt of the convex-face proof.

The two-interval arithmetic kernel is covered by Li–Chen's uniform bound for continued fractions of ratios of parameter polynomials [7, Theorem 27]. For several intervals, finite residue branches and a uniform compressed node count, the stored survey did not locate an identical general theorem; this is not a novelty proof.

K3's variable-modulus behaviour obstructs a single ordinary multivariable Presburger description in fixed common coordinates. Moving to canonical representatives or allowing variable divisibility does not follow from raywise periodicity. The stronger M4/M4-prime questions remain separate. General E10 $r = 1$ necessity, untranscribed families, all degenerations, human proof peer review, and a formal geometric-to-program pipeline are not completed here. A3 minimum, density and the IPR census belong to the first article. No new D12 classification is asserted by this paper.

Appendix A. Local polygons and zero boundaries

Machine-readable claim: `app:domains`.

Twice total polygon area is the carbon order only when the quotient is valid. The stored local coefficient audit gives

Signature	Side factors	Twice total area
E10	$p, r, r+s$	$12r^2 + 2s^2 + 12rs + 24pr + 12ps$
D12	$s, r+s$	$12r^2 + 48rs + 48s^2$
D14	r, s	$8r^2 + 24rs + 12s^2$
D1	p, r	$36p^2 + 48pr + 12r^2$
K3	r, s	$14r^2 + 12rs$
B2	p, r	$12p^2 + 24pr + 4r^2$
C1	a, b	$8a^2 + 16ab + 4b^2$
O2	t, d	$14t^2 + 16td + 2d^2$

The 71-chart/342-edge audit establishes local convexity, positive area and side lengths in its stated positive domain, not global quotient simplicity. With other parameters positive, E10 $s = 0$ and D12 $r = 0$ alone remain locally nondegenerate. E10 $p = 0$ or $r = 0$, K3 $r = 0$, B2 $p = 0$ or $r = 0$, C1 $a = 0$, and O2 $t = 0$ collapse some faces. D12 $s = 0$, either D14/D1 zero, K3 $s = 0$, C1 $b = 0$ and O2 $d = 0$ produce zero sides. Removing an IET zero-length label is not the same as redefining a graph after cone mergers. The omitted leapfrog-domain data remain in the extension's prospective records, not the main corollary. A3's $l = 0$ and $k = l$ are the separate A1 and A2 boundaries.

Appendix B. Representative formal statements and source correspondence

The accompanying `v3-correspondence.md` maps every numbered v3 section to this reorganization, including the moved oblique-direction sketch. It distinguishes translation, condensation and formal-status updates; no prose-equivalence theorem is claimed.

The primary received declarations are `Claude2.k3_ray_all`, `Claude2.e10_ray_all`, `Claude2.k3_example`, `Claude2.e10_example`, and `Claude2K3.k3_sufficient`. The machine layer supplies their exact types, source commit and the kind of proof receipt. The historical representative kernel types remain:

> `finite_literal_path_decision`: finite type, two-common-neighbour hypothesis, cone predicate and fixed oriented triangle imply a first repeat at most the number of vertices, and path existence iff a legal endpoint occurs before it.

`cylinder_portal_shift`: a Bézout witness for primitive u and $g \mid Q/\mathrm{cylHeight}(u, v)$ imply that the stated shifted displacement is a multiple of gu. The integer-division and divisibility hypotheses are retained.

`induction_uniform_depth`: a finite-list child relation with a strictly decreasing lexicographic natural rank has a finite uniform depth from each starting state.

`iet_component_uniform_depth`: the concrete equal-final-label removal component has such a depth; the omitted IET branches are not thereby formalized.

The supplied exact sources, not these paraphrases, determine the Lean types. Dependencies are pinned; caches, binaries, raw graph streams, contact details are supplied separately from the proof sources.

Acknowledgements and AI-use statement

The author used Codex and Claude Code to assist with literature organization, mathematical exploration, proof drafting, implementation, finite verification, formalization of the explicitly identified arithmetic and graph statements, and English exposition. The author is responsible for the published account, its provenance, and the stated limits of verification. AI systems are not listed as authors. AI prose reviews are identified separately from human review. Only the explicitly listed Lean propositions are described as formally proved; finite checks and independently constructed tests have their stated domains. No claim of complete human checking of every argument is made.

License and contact

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